Lean Six Sigma Practice Test W Answers

1. The purpose of Six Sigma is to:

   A. Fix defects.
   B. Reduce variation.
   C. Perform project management.
   D. Calculate the standard deviation.

Explanation:
The purpose of Six Sigma is to reduce variation from any process and product. As variation is reduced, the quality is increased.

2. What is Six Sigma?

   A. A business philosophy focusing on improvement
   B. A project management model
   C. A defects reduction technique that is six times better than average
   D. A measurement system

Explanation:
Six Sigma is a business philosophy focused on driving continuous and breakthrough improvement.

3. DMAIC stands for:

   A. Design, Measure, Analyze, Improve, Control
   B. Define, Measure, Analyze, Improve, Control
   C. Double Measurement And Instructions Checklist
   D. Definition, Measurements, Analysis, Improvements, and Control

Explanation:
DMAIC is a Six Sigma problem-solving methodology that stands for Define, Measure, Analyze, Improve, and Control.

4. DFSS stands for:

   A. Design and Fix using Six Sigma
   B. Develop Fast using Six Sigma
   C. Define Futuristic Service and Solutions
   D. Design For Six Sigma

Explanation:
DFSS is Six Sigma methodology that stands for "Design for Six Sigma" used while designing new processes.

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5. Six Sigma Level can be achieved by having:

A. Less than 3.4 defects per million opportunities
B. Six times better quality than competitors
C. Six levels of testing and validation before release
D. Six Nines Quality (99.9999%)

Explanation:

If your process is delivering less than 3.4 defects per million opportunities, then you can call your process as performing at Six Sigma level.

6. DMADV stands for:

A. Design, Measure, Analyze, Develop, Verify
B. Develop Measurement and Analysis Department for Verification
C. Define, Measure, Analyze, Design, Verify
D. Design, Measure, Analyze, Define, Validate

Explanation:

DMADV is Six Sigma methodology that is used when developing a new process. It stands for "Define, Measure, Analyze, Design, and Verify."

7. The purpose of the Toll Gate review is to:

A. Confirm if the earlier phase was completed successfully.
B. Get a formal sign-off from the stakeholders.
C. Review progress and kick off the next phase.
D. All of the above

Explanation:

It is all of the above. The purpose of the Tollgate is to ensure plans are completed successfully, acquire formal sign-off from stakeholders and then review the progress and kick off the next phase.

8. What is the primary role of Master Black Belt?

A. Provide a budget for the projects
B. Coach, mentor, and train Black Belts
C. Be a master in Six Sigma implementation
D. Review projects

Explanation:

The primary role of Master Black Belt is to provide guidance, train, coach, and mentor Black Belts for Six Sigma projects.

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9. Which role performs the following responsibilities: "Evangelize Six Sigma, provide resources and resolve any cross-functional issues"?

A. Sponsor
B. Senior Executive
C. Master Black Belt
D. Champion

Explanation:

It is the Champion that is responsible for evangelizing Six Sigma and providing resources and support for Six Sigma projects

10. A Black Belt is expected to deliver what types of results from Six Sigma projects?

A. At least 200,000 USD as an annual benefit
B. Have everybody in the organization trained on Six Sigma
C. Large projects
D. Defects reduction projects

Explanation:

Six Sigma Black Belt level projects are expected to provide returns of more than 200,000 USD annually.

11. The purpose of capturing Voice of Customer is to:

A. Ensure there is no confusion between the contract and delivery
B. Gather customer feedback
C. Have a sampling exercise
D. Capture stated or unstated customer needs or requirements

Explanation:

The VOC mechanism is used to capture stated or unstated customer needs or requirements.

12. In which role is a full-time professional to lead large Six Sigma projects?

A. Six Sigma Champion
B. Green Belt
C. Master Black Belt
D. Black Belt

Explanation:

The Black Belt is typically a full-time role that is responsible for leading enterprise-wide large scale projects and mentoring Green Belts.

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13. What is the role of Black Belts in dealing with organizational roadblocks?

A. Escalating issues to Senior Management
B. Communicating the benefits of Six Sigma, changing efforts and building confidence in the team
C. Asking for help from Master Black Belts to handle a situation
D. Providing Six Sigma training to everybody in the organization

Explanation:

The Black Belt is expected to work closely with Master Black Belt and Senior Management to communicate the benefits of Six Sigma, change efforts, and build confidence in the team.

14. Why do Black Belts and Master Black Belts conduct closed-looped assessments of Six Sigma implementation?

A. To ensure all aspects of Six Sigma have been implemented
B. To assure the privacy of the people involved in Six Sigma projects
C. To ensure competitors do not get the information on the improvements achieved
D. To minimize influencing people with the Six Sigma project measurements and avoid any manipulation

Explanation:

The Six Sigma Black Belt and Master Black Belt usually conduct the assessment in a closed loop to make sure that all aspects of Six Sigma have been implemented in the project.

15. What tool did Kaoru Ishikawa develop?

A. SWAT Analysis Tool
B. CTQ Diagram
C. Cause and Effect Diagram
D. FMEA Tool

Explanation:

Ishikawa developed the Cause-and-Effect diagram, also known as Ishikawa tool or Fishbone diagram.

16. What is the best mechanism to collect VOC data?

A. Complaints Database
B. Customer Support History
C. Industry Reports
D. Surveys and Questionnaire

Explanation:

It is always good to collect data directly from the customer, and surveys and questionnaires are a good way to collect these data.

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17. Which of the following Six Sigma tools can be used by a team to prioritize a list of options to improve?

A. Brainstorming
B. Nominal Group Technique
C. Cause and Effect Diagram
D. 5 Whys

Explanation:
After the group brainstorms ideas, they can get them prioritized using the Nominal Group Technique.

18. What does SWOT stand for?

A. Strengths, Weaknesses, Opportunities, Threats
B. Six Work Optimization Techniques
C. Seven Workplace Opportunities for Transformation
D. Strengths, Weaknesses, Opportunities, Techniques

Explanation:
SWOT stands for Strengths, Weakness, Opportunities, and Threats.

19. The primary factor in the successful implementation of Six Sigma is to have:

A. Training for everybody in the organization
B. Support from everybody in the organization
C. Experienced Six Sigma practitioners
D. Executive commitment and support

Explanation:
Without executive commitment and support for a Six Sigma project, it would not be possible for anybody with or without Six Sigma experience to implement it successfully.

20. What is Continual Improvement Process?

A. The principle of never stopping any improvements
B. Improvements being made while people and machines continue to work
C. A steady effort made to improve existing processes, products, or services
D. The non-stop cyclic process of improvements where one improvement leads to another

Explanation:
Continual Improvement Process or CIP is a steady effort made to improve existing processes, products, or services in continuous, regular intervals or discrete jumps.

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21. What type of chart one would typically use for analyzing positional, cyclical, and temporal variations?

A. Multi-Vari Charts
B. SPC Charts
C. Cause and Effect Diagram
D. Run Charts

Explanation:

When multiple data points related to positional, cyclical, and temporal variations are to be analyzed, multi-vari charts are most appropriate to be used.

22. If you need to collect data to analyze the symptoms of a problem, what tool will be most appropriate?

A. Flow Chart
B. FMEA Tool
C. Force Field Analysis
D. Check Sheet

Explanation:

To collect data related to any process, always start with a Check Sheet. It is a good technique to capture problem symptoms.

23. The purpose of having functional requirements is:

A. To describe a single, measurable performance
B. To provide details on how a product or service needs to operate
C. To provide specification limits
D. To build VoC

Explanation:

The purpose of having functional requirements is to provide details on how a product or service needs to operate.

24. Increasing performance in a Six Sigma project from 3 sigma to 4 sigma would reduce defects per million by a factor of:

A. 2
B. 8
C. 10
D. 16

Explanation:

At 3 sigma level Defects per million opportunities is 66,807. Whereas at 4 sigma level it is at 6,210 defects per million. The factor reduction in the defect is 10.758 which is rounded to 10.

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25. In a Cause and Effect diagram, some of the major cause categories are mother nature, management, measurement systems, and the following four standard causes:

A. Marketing, methods, material, machines
B. Man, material, methods, machines
C. Man, manufacturing, methods, material
D. Man, material, medicines, machines

Explanation:

The major cause categories in Cause and Effect diagrams are man, material, methods, and machines, along with mother nature, management, and measurement systems.

26. To monitor the number of defects on different sample sizes, which of the following control charts is most appropriate to use?

A. np chart
B. c chart
C. u chart
D. p chart

Explanation:

The control chart "u chart" is most appropriate to monitor the number of defects in different sample sizes.

27. What are the 5 elements of Kaizen?

A. Teamwork, Personal Discipline, Improve Morale, Quality Circles, Suggestions for Improvement
B. Define, Measure, Analyze, Improve and Control
C. Define, Measure, Analyze, Design and Validate
D. Identify Value, Map the Value Stream, Create Flow, Establish Pull, Seek Perfection

Explanation:

The 5 elements of Kaizen are Teamwork, Personal Discipline, Improve Morale, Quality Circles, and Suggestions for Improvement

28. What are the 5 Lean principles?

A. Identify Value, Map the Value Stream, Create Flow, Establish Pull, Seek Perfection
B. Teamwork, Personal Discipline, Improve Morale, Quality Circles, Suggestions for Improvement
C. Define, Measure, Analyze, Improve, and Control
D. Reduce Waste, Organize, Just-In-Time, optimize, Improve

Explanation:

The 5 principles of Lean are Identify Value, Map the Value Stream, Create Flow, Establish Pull, and Seek Perfection.

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29. Measurement accuracy is attained when:

A. Different people get the same result when measuring the same item or characteristic
B. The same person taking multiple measurements on the same item or characteristic gets the same results every time
C. The measured value has little deviation from the actual value
D. The resolution of the measurement instrument can give at least 5 distinct values in the range being measured

Explanation:

Repeatability will help reduce variation between different measurements and similarly, reproducibility will help reduce variations across measurements by different people. Both of these will help, and true Measurement Accuracy will be achieved when the difference or deviation between the measured value and actual value is reduced.

30. In a Normal distribution, approximately 68% of the data will occur within:

A. +/- 2 standard deviation
B. +/- 1 standard deviation
C. +/- 3 standard deviation
D. Outside the bell-shaped curve

Explanation:

In a normal distribution, the area under +/- 1 standard deviation covers approx. 68% of the data.

31. In FMEA exercise, the Risk Priority Number (RPN) is calculated as:

A. Severity x occurrence x risk
B. Severity x risk x detection
C. Occurrence x detection x control
D. Severity x occurrence x detection

Explanation:

Risk Priority Number is generated by calculating product of severity, occurrence, and detection.

32. Which of the following activities is a value-added activity?

A. Testing and Inspection
B. Setup
C. Process
D. Storage

Explanation:

Any time and effort spent on testing, inspection, storage, and setup is a non-value added activity. The only activity that is adding any value to the product is 'processes.

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33. What does Lean help you with?

A. Reducing weight
B. Eliminating waste and reducing non-value adding activities
C. Achieve higher quality
D. Improving employee morale

Explanation:
Lean helps you understand what are value-add and non-value-add activities and helps you reduce the latter and eliminate waste.

34. We want to test the performance of people before and after training. Which of the following tests can be used if the training improved workers Performance?

A. 2-sample z test
B. Null hypothesis
C. Paired t-test

Explanation:
The Paired t-test is used to determine performance before and after training.

35. Which of the following is a chart that depicts the frequencies of measurement data?

A. Histogram
B. Run Charts
C. Control charts
D. Pie Chart

Explanation:
The Bar Chart type histogram is typically used to depict the frequency of occurrence of each of the measurement data.

36. The purpose of using control charts is to:

A. Detect the root causes of defects
B. Validate Measurements System
C. Control the output of a process
D. Evaluate process performance over time

Explanation:
The purpose of the Control Chart is to show visually the expected performance of any process and set limits. It provides details about process performance over a period of time, which can be evaluated further as needed.

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37. Design for Test is a design technique that offers a benefit of:

A. Customer Satisfaction Tests
B. Defect Diagnosis
C. Manufacturability
D. Smaller Process Variation

Explanation:

Design for Test as a design technique that aims to introduce a testing capability in the manufacture of electronics. The goal of this technique is to introduce and validate the capability of diagnosing product defects.

38. What does DFMEA stand for?

A. Design Failure Mode and Effect Analysis
B. Defect Failure Mode and Effect Analysis
C. Derive Failure Mode and Estimation Analysis
D. Design Failure Mode and Estimation Analysis

Explanation:

DFMEA (Design Failure Mode and Effect Analysis) is the application of the Failure Mode and Effects Analysis (FMEA) method specifically for product/service design.

39. Which chart is used to prioritize the highest overall score received as part of the CE matrix?

A. Glaphira Charts
B. Standard Bar Charts
C. Putoro Charts
D. Pareto Charts

Explanation:

A Pareto chart is used to graphically summarize and display the relative importance of the differences between groups of data.

40. Which of the following is a basic pre-condition for Six Sigma to succeed?

A. Qualified resources
B. Well-defined process
C. Trusted suppliers
D. Reliable funds

Explanation:

To begin a Six Sigma project, ensure that you have a well-defined process. If such a process does not exist, it is advisable to first document the process and then start the project.

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41. What does SIPOC stand for?

A. Stakeholders, Inputs, Process, Outputs, and Clients
B. Stakeholders, Inputs, Process, Outputs, and Customers
C. Suppliers, Inputs, Process, Outputs, and Customers
D. Suppliers, Inputs, Purchase, Outputs, and Customers

Explanation:

An SIPOC diagram is a tool used by a team to identify all relevant elements of a process improvement project before work begins. It helps define a complex project that may not be well scoped and is typically employed at the Measure phase of the Six Sigma DMAIC (Define, Measure, Analyze, Improve, and Control). Methodology.

42. Which question needs to be asked before preparing the SIPOC?

A. Who is the stakeholder for the process?
B. What are the inputs?
C. What resources does the process use?
D. All of the above

Explanation:

For preparing SIPOC, we need to have answers to who the supplier, inputs, customers, process, and outputs are.

43. What are the various types of measurement scales?

A. Nominal, Ordinal, Interval, and Ratio
B. Nominal, Ordinal, Inverse, and Proportional
C. Direct, Indirect, Inverse, and Proportional
D. Regular, Variance, Inverse, and Proportional

Explanation:

The key measurement scales are Interval, Nominal, Ordinal, and Ratio.

44. What does nominal measurement define or classify?

A. Information about inputs from stakeholders
B. Set of mandatory processes
C. Presence or absence of a certain attribute
D. Minimal length of the project

Explanation:

Nominal scales define or classify only the presence or absence of a certain attribute. With nominal scales, you can only count items and the statistics that you can use are Percentage, Proportion and Chi-Square tests.

45. Which measurement scale is a comparative scale of management?

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A. Ratio
B. Interval
C. Variance
D. Ordinal

Explanation:

An ordinal scale defines one item having more or less of the attribute than the other item. In simple words, it is a comparative scale of measurement. Example --- Taste and attractiveness. As statistical measures, Rank order correlation is typically used.

46. What does Ratio Scale define?

A. The ratio of supply vs. demand
B. The ratio of supporters vs. resistors
C. All the true zero value points
D. Only the positive measurements

Explanation:

The Ratio Scale defines all true zero value points. You can perform addition, subtraction, division, and multiplication with such data points. Example --- Elapsed time, distance, etc.

47. Data can be categorized into which of these divisions?

A. Continuous or Variable
B. Discrete
C. Attribute
D. All of the above

Explanation:

We can categorize data in three divisions --- Continuous or Variable, Discrete, and Attribute Data.

48. Which is the most popular and most often used measure of Central Tendency in Six Sigma applications?

A. Mode
B. Mean
C. Median
D. Variance

Explanation:

Mean is the arithmetic average of all the data points and is popularly known as Arithmetic Mean. This is the most popular and most often used measure of Central Tendency in Six Sigma applications.

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49. Which measure of Central Tendency in Six Sigma applications should be used when the data has outliners?

A. Mode
B. Mean
C. Median
D. Variance

Explanation:

Median, also known as Positional Mean, is the central location or position of the data list. Mean gets influenced by the outliers. Median doesn't change much even with the introduction of Outliers.

50. What are the measures of dispersion?

A. Mean, Mode, and Median
B. Range, Variation, and Standard Deviation
C. Mode, Medium, and Range
D. Median , Range, and Variation

Explanation:

The measures of dispersion are Range, Variation, and Standard Deviation. The range is the difference between the maximum value and the minimum value. Variance is the Average of Squared Mean Differences. Standard Deviation is the Square root of Variance.

51. By which tool can measurement system accuracy be verified?

A. MSA or GAGE RR
B. CSA or GAGE RR
C. MSA or RAGE RR
D. CSA or RAGE RR

Explanation:

MSA or GAGE RR is a tool that is used to validate the measurement system to see if the measurement system is able to give us data that can be relied on.

52. What are the different categories in which measurement system errors can be classified?

A. Accuracy and Precision
B. Clarity, Accuracy, and Precision
C. Resolution, Accuracy, and Precision
D. Clarity and Resolution

Explanation:

Measurement system errors fall into 3 major categories: Resolution: The ability to provide granular readings Accuracy: Difference between observed measurement and actual measurement.

Precision: Difference observed when measuring the same part with the same equipment or instrument.

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53. What are the 3 components associated with measurement systems accuracy?

A. Repeatability, Bias, and Linearity
B. Stability, Bias, and Linearity
C. Repeatability, Reproducibility, and Linearity
D. Stability, Bias, and Reproducibility

Explanation:

Measurement systems accuracy consists of Stability: Consistency and predictability of measurements over time Bias: Change in the actual value with respect to reference value Linearity: A measure of bias values through a range of measurements

54. What are the 2 components associated with measurement systems precision?

A. Stability and Bias
B. Bias and Linearity
C. Repeatability and Reproducibility
D. Stability and Reproducibility

Explanation:

Precision deals with repeatability and reproducibility. Repeatability is also known as Equipment Variation. Reproducibility is also known as Appraiser Variation.

55. Which one of these is not a property of a good measurement system?

A. Stability
B. Repeatability
C. Bias
D. Linearity

Explanation:

A good measurement system should possess the following properties or characteristics: Accuracy, Repeatability, Linearity, Reproducibility, and Stability. Bias is not one of the properties.

56. A standard with a known value of 25.4 mm is checked 10 times by one mechanical inspector using a dial caliper with a resolution of 0.025 mm. The readings obtained are: 25.425, 25.425, 25.400, 25.400, 25.375, 25.400, 25.425, 25.400, 25.425, 25.375. The tolerance value of this gage is ± 25 mm. Calculate the Bias value and % Bias.

A. 0.05, 1%
B. 0.01, 0,2%
C. 0.1, 10%
D. 0.5, 50%

Explanation:

1) Find the Average of Measurements. In this case, Xbar = 25.405 mm 2) Bias = Average - Reference Value = 25.405 - 25.4 mm = 0.05 mm 3) % Bias = 100 * (Bias/Tolerance) % = 100 * (0.05/0.5)

% = 1%

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57. Which chart indicates Part Variation?

A. Puerto Chart
B. Glaphira Chart
C. Pareto Chart
D. Xbar/Averages Chart

Explanation:

An Xbar chart is used to observe and evaluate the behavior of a process over time and take corrective action if necessary. The chart plots the average values of a number of small sampled subgroups.

58. Which formula is used to measure Total Process Variation (?t) in terms of Gage(?m) and Part(?p) Variation?

A. (σm + σp)/2
B. √(σm2+ σp2)
C. (σm + σp)
D. ((σm + σp) * 2)1/2

Explanation:

σm = Gage Variation σp = Part Variation σt (Total Process Variation) = √(σm2+ σp2)

59. What is Gage Variation (?m) in terms of Repeatability(?r) and Reproducibility(?o) variation?

A. (σr + σo)/2
B. (σr + σo)
C. (σr squared+ σo squared)1/2
D. ((σr + σo)/2))1/2

Explanation:

σm = √(σr2+ σo2) where σr = Repeatability or Equipment Variation σo = Reproducibility or Appraiser Variation

60. What is the percentage threshold for GRR above which the gage needs to be replaced?

A. 0.25
B. 0.3
C. 0.35
D. 0.4

Explanation:

% GRR = 100 * σm /Tolerance % % PV = 100 * σp /Tolerance % ndc = 1.41 * PV/GRR If

%GRR > 30%, gage needs to be replaced.

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61. While selecting a sample, what rules should govern your choice?

A. Any sample size, if randomly selected, can be suitable for analysis. Since it is sampling, it is not expected to have a specific number of samples.
B. 10% of the total
C. One sample every hour
D. Determine sample size using formula and base it on the tolerance and confidence interval expected and standard deviation or the current error rate of the population

Explanation:

The sample size has to be determined statistically using formulas that take into consideration standard deviation or current error rate of the population.

62. Who is the process expert that does the rating of the samples, i.e., as okay, defective, etc.?

A. Master Champion
B. Master Appraiser
C. Process Champion
D. Process Appraiser

Explanation:

The process expert who does the rating of the samples is called a Master Appraiser.

63. What does SPC stand for?

A. Statistical Process Control
B. Standard Process Control
C. Statistical Process Chart
D. Standard Process Chart

Explanation:

As per SPC or Statistical Process Control, every measurable phenomenon is a statistical distribution, i.e. every set of observed data is a set of effects of unknown common causes.

64. What does SEM stand for?

A. Standard Error of Mean
B. Standard Error of Mode
C. Standard Estimation of Mode
D. Standard Estimation of Mean

Explanation:

The standard deviation of the averages denoted by SEM or Standard Error of Mean is represented by the standard deviation of population divided by the square root of sample size.

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65. Which of the following is not a capability index for continuous data?

A. Process Capability
B. Process Capability Index
C. Process Performance
D. Process Performance Index

Explanation:
Process Performance is not a measurement for capability index for continuous data.

66. What is the formula for calculating Process Capability?

A. (USL-LSL)/(6 * σ)
B. (USL+LSL)/(6 * σ)
C. (USL+LSL)/(σ)
D. (USL+LSL)/(2*σ)

Explanation:
Cp = (USL-LSL)/(6 * σ), with USL – LSL representing the Process Tolerance.

67. How do you derive the Process Capability Index (Cpk) once you have the Cpl and Cpu values?

A. min (Cpl, Cpu)
B. max (Cpl, Cpu)
C. avg (Cpl, Cpu)
D. stddev (Cpl, Cpu)

Explanation:
The Process Capability Index Cpk is the minimum of both Cpl and Cpu.

68. When is the Mean not centered?

A. Cpk > Cp
B. Cpk = Cp
C. Cpk < Cp
D. None of the above

Explanation:
If Cpk < Cp, mean is not centered. If Cpk = Cp, mean is centered, and the process is accurate.

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69. Which one of these activities should ideally come first in the implementation of a Measure phase?

A. Checking Normality
B. Checking Precision
C. Checking Repeatability
D. Checking Resolution

Explanation:
Checking Resolution is the first activity in a Measure Stage.

70. In a typical Measurement System Analysis activity, the GRR is 42% and the contribution is split equally among Gage Variation and Part Variation. What would you do?

A. Leave the GAGE. It is acceptable
B. Calibrate the equipment
C. Calibrate the equipment and check operator variation
D. Calibrate the equipment, check operator variation, and identify Part Variation

Explanation:
Presence of Part Variation in the study indicates that it needs to be studied.

71. The tool that you typically use in the Measure stage to scope a project is:

A. DFMEA
B. PFMEA
C. CE Matrix
D. SIPOC

Explanation:
SIPOC is the best possible scoping tool for the project.

72. Which one of these is the most important precursor for calculating Capability?

A. Process Stability
B. Normality
C. No skew in data
D. None of the above

Explanation:
Process stability is the most important task to be done before any calculations are done in Measure.

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73. Which of these measures would you normally associate with Process Capability of Discrete Data?

A. DPMO/PPM
B. DPU
C. Cp
D. Cpk

Explanation:
With Discrete Data, DPMO and PPM are the best possible metrics showing capability.

74. Which of the following methods directly helps you determine batch size?

A. Setup time
B. Changeover time
C. Cycle time
D. EPEI

Explanation:
Every Part Every Interval (EPEI) is the method for determining the batch size.

75. Which type of chart is appropriate when sample sizes are variable and each sample may contain more than one instance of the targeted condition?

A. P chart
B. Autocorrelation chart
C. U chart
D. X-bar chart

Explanation:
A U chart is appropriate when sample sizes are variable and each sample may contain more than one instance of the targeted condition. These are control charts most appropriate for handling attributes data.

76. What is the purpose of PERT analysis during the Analyze phase of DMAIC?

A. To identify the most influential steps in a process
B. To monitor improvements in cycle time
C. To identify the critical path of a process
D. To reduce cycle time

Explanation:
The purpose of program evaluation and review techniques (PERT) analysis during the analyze phase of DMAIC is to identify the critical path of a process.

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77. Which form of leveling typically includes the use of heijunka boxes?

A. Leveling by volume
B. Leveling by product
C. Demand leveling
D. Customer leveling

Explanation:

Leveling by-product typically includes the use of heijunka boxes. A heijunka box is a tool for scheduling in which the tasks to be completed and the estimated times for initiation and completion are listed. The point of leveling by-product is to reduce inefficiency by adjusting the mix of products.

78. While performing monthly reviews, in the control chart of one of your production unit, you observe that the last 50 points are very near to the centerline. All of them appear to be within about one sigma of the centerline. What Interpretation would you derive from this?

A. Somebody miscalculated the original calculations of the control limits
B. The process standard deviation has decreased during last month and hence the last 50 data points are showing values much closer to the centerline, But, nobody thought to recompute the control limits to reflect the new reality
C. This is an out of control situation. Get a team in place right away and see what the trouble is.
D. The process is in control. The closer the points are to the centerline the better control of the process is. No action is needed, process and the control limits to continue as it is

Explanation:

Answer A is obviously incorrect, the control limits would have been set up correctly in the first place. Answer C is incorrect, the process is not out of control. Answer D is partially correct because as the process improves, we need to fix the limits as well. So Answer B is the correct answer.

79. In an analysis of variance, how is the F statistic used?

A. To compare the mean square treatment with the mean square error
B. To estimate the process average
C. To find the variation within each subgroup
D. To find the variation between different subgroups

Explanation:

In an analysis of variance, the F statistic is used to compare the mean square treatment with the mean square error. The mean square treatment is the average variation between the subsets, while the mean square error is the sum of the squares of the residuals.

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80. The flow of activities that include value-added as well as non-value added activities which take the raw material and deliver the end product to the customer is called:

A. Value Stream Map
B. Process Analysis
C. SIPOC
D. Non Value Add Activities Analysis

Explanation:

A Value Stream has a flow of activities that consist of value-added as well as non-value-added activities which take the raw material and deliver the end product to the customer. Value Stream Map is a visual representation of the same.

81. Which of the following are Warusa Kagen conditions?

A. Meda, Mera, and Meri
B. Mauda, Maura, and Mauri
C. Muda, Mura, and Muri
D. Moda, Mora, and Mori

Explanation:

Muda, Mura, and Muri are Warusa Kagen conditions.

82. Tremendous resistance can be expected in which of the Value Stream Analysis Phase(s)?

A. Mura
B. Muda
C. Muri
D. All of the above

Explanation:

When analyzing the value stream, the Black Belt needs to first identify the Muda in the system. In typical cases, Muda accounts for 90% of the time spent in operations.

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83. What does CLOSEDMITTS stand for?

A. Complexity, Labor, Overproduction, Space, Energy, Defects, Materials, Idle Time, Time, Transportation, Safety Hazards
B. Complexity, Labor, Overproduction, Space, energy, Design, Materials, Innovation, Time, Transportation, Safety Hazards
C. Client, Labor, Overproduction, Stakeholders, Energy, Design, Materials, Innovation, Time, Transportation, Safety Hazards
D. Client, Labor, Optimization, Stakeholders, Energy, Design, Materials, Innovation, Time, Transportation, Safety Measures

Explanation:

C - Complexity - Unnecessary steps and excessive documentation L - Labor - Inefficient operations and excess headcount O - Overproduction - Producing more or producing before than customer needs S - Space - Storage space for inventory E - Energy - Wasted human energy D - Defects - Repair or Rework in products M - Materials - Scrap or ordering more than needed I - Idle time - Material sits for time T - Time - Time waste T - Transportation - Movement adding no value S - Safety hazards - Unsafe environments

84. Spaghetti charts graphically depict the movement of and in a process.

A. products and people
B. resources and labor
C. income and expenses
D. funds and allocations

Explanation:

The Spaghetti Chart helps understand the Physical Process Map and not necessarily just the Value Map. These charts graphically depict the movement of products and people in a process.

85. What does CE diagram stand for?

A. Customer and Engagement
B. Cause and Estimation
C. Client and Expectation
D. Cause and Effect

Explanation:

CE Diagram or Cause and Effect Diagram, also known as the Fishbone Diagram, is a very popular and an important tool that gives you a list of all the issues due to which a problem occurs.

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86. Box Plot is also known as:

A. Whiskers Plot
B. Candlestick Plot
C. Box and Whiskers Plot
D. All of the above

Explanation:

Box Plot is another useful tool that helps you to know the nature of variability in a process. It is also known as Whiskers Plot, Candlestick Plot, and Box and Whiskers Plot.

87. Which kind of information does Box plot provide?

A. Location, spread, and variability of data
B. Distance, relativity, and density of data
C. Location, speed, and density of data
D. Continuity, relativity, and variability of data

Explanation:

These plots are often used in the Analyze phase to understand the process variability. They help you see the distribution of values in several groups. Box plots also tell you information about location, spread, and variability of data.

88. What does KPIV stand for?

A. Key Process Indicator Value
B. Key Person Indicator Value
C. Key Process Input Value
D. Key Person Input Value

Explanation:

KPIV stands for Key Process Input Value. Typically, the Fishbone Diagram would give an insight into the possible KPIVs for a model.

89. To design for manufacturability, the following methods are introduced, except:

A. Standardization and simplification
B. Fewer active parts are used
C. Design alternative are used
D. Product quality is held above simplicity

Explanation:

Design for manufacturability means that products are standardized and simplified so that there are fewer products with defects made in every production cycle. Product quality is held above simplicity is not introduced, hence, this is the right choice for the question

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90. A limitation in Six Sigma implementation is when management applies the methodology:

A. Using trained professionals
B. Using approved funds
C. Only for a certain fixed duration
D. In a few departments within the company

Explanation:

It is important that the methodology is applied in all the departments of the company and not limited to any specific groups. This is required to ensure that there are no blockages at a later stage and fruitful/positive results are seen as an output across the organization.

91. In a scenario where one response variable and one predictor variable are to be tested, which regression analysis needs to be used?

A. Aggregated Linear Regression
B. Single Linear Regression
C. Sample Linear Regression
D. Simple Linear Regression

Explanation:

Simple Linear Regression is to be used when you have one response variable (KPOV) and one predictor variable (KPIV) to test.

92. When is Multiple Linear Regression to be used?

A. 1 KPOV and 1 KPIV
B. 1 KPOV and 2 KPIV
C. 2 KPOV and 1 KPIV
D. 2 KPOV and 2 KPIV

Explanation:

Multiple Linear Regression is to be used when you have one response variable (KPOV) and two multiple predictor variables (KPIV) to test.

93. _____ shows the extent to which multi-collinearity exists between predictors.

A. Multiple Linear Regression
B. Variation Inflation Factor
C. Single Linear Regression
D. Statistical Accuracy

Explanation:

VIF (Variance Inflation Factor). VIF shows the extent to which multi-collinearity exists amongst predictors.

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94. Which of the following tests for the presence of autocorrelation between residuals?

A. Duminy Walker Statistic
B. Durbin Walker Statistic
C. Durbin Watson Statistic
D. Duminy Watson Statistic

Explanation:
Durbin Watson Statistic Tests are used to determine the presence of autocorrelation in residuals.

95. What does PRESS in PRESS Statistic stand for?

A. Partial Residuals Sum of Squares
B. Part Regression Six Sigma
C. Predicted Residuals Six Sigma
D. Predicted Residuals Sum of Squares

Explanation:
PRESS stands for Predicted Residuals Sum of Squares in PRESS Statistic.

96. Which regression procedures are used in stepwise regression?

A. Standard Stepwise, Forward Selection, and Backward Elimination
B. Standard Stepwise, Forward Estimation, and Backward Estimation
C. Standard Stepwise, Forward Regression, and Backward Regression
D. Standard Stepwise, Forward co-relation, and Backward co-relation

Explanation:
In Stepwise Regression, the most significant variable is added or removed from the regression model. Three common Stepwise Regression procedures are Standard Stepwise, Forward Selection, and Backward Elimination. Use of Mallow's Cp Statistic in Stepwise Regression is popular.

97. What is Confidence Interval?

A. The interval between 2 consecutive sets of data population
B. An estimated range of values which is likely to include the parameter of the population
C. The interval between each measurement to gain confidence
D. The interval between the Measure and Analyze phase

Explanation:
Confidence intervals give an estimated range of values, known as Margins or Confidence Interval widths, which are likely to include the parameter of the population.

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98. Which factors impact the confidence intervals for sample data?

A. Data Variation and Data Continuity
B. Accuracy and Precision
C. Significance Level and Sample Size
D. Data Relativity

Explanation:

Significance level and sample size impact the confidence interval - The choice of ? for the confidence intervals dictates the margin of error or confidence interval width. With an increase in sample size at the same significance level, you will find that the confidence intervals width keeps decreasing.

99. Which statistic can be used to determine the probability of variances occurring?

A. Binomial Statistic
B. Poisson Statistic
C. Moving Average Statistic
D. Chi-Square Statistic

Explanation:

The Chi-Square statistic can be used to determine the probability of variances occurring. $\chi 2 = (n-1)s2/\sigma 2$

100. For a population variance of 1 and sample variance on data collected of 0.6, determine the probability that the sample variance can exceed 0.6. The sample size is 30.

A. A. 15.4
B. B. 16.4
C. C. 17.4
D. D. 18.4

Explanation:

$\chi 2 = (n-1)s2/\sigma 2 = (30-1)0.6/1 = 29 * 0.6 = 17.4$

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